KAPPLER Thomas BTS2

Déploiement d'un service avec Docker : Debian12-Apache2

Préparer le contexte :

Crée un dossier de travail :

```
root@docker:/var/lib/docker/volumes/public_html/_data# mkdir -p ~/contexte/debian12-apache2
```

Crée ta page web personnalisée :

```
root@docker:~/contexte/debian12-apache2# nano index.html
```

```
<html>
 <head>
<title>Apache Debian 12 - Thomas KA</title>
</head>
 <body>
<h1>Test Apache2 sur Debian 12</h1>
 <p>Image Docker de thomaska13.</p>
 <p>TP Docker : Déploiement d'un service</p>
 </body>
</html>
```

Créer le Dockerfile :

Crée le fichier docker :

```
root@docker:~/contexte/debian12-apache2# nano dockerfile
```

```
FROM debian:12

LABEL maintainer="thomaska"
LABEL description="Debian 12 avec Apache2 et page web statique"

# Variables d'environnement pour les locales
ENV LANG=fr_FR.UTF-8

# Installation d'Apache2, tzdata et locales
RUN apt update \
    && apt -y --no-install-recommends install \
        apache2 \
        tzdata \
        locales \
    && rm -rf /var/lib/apt/lists/* \
    && apt clean

# Configuration timezone Europe/Paris et locales FR
RUN rm /etc/localtime \
    && ln -s /usr/share/zoneinfo/Europe/Paris /etc/localtime \
    && dpkg-reconfigure --frontend=noninteractive tzdata \
    && sed -i 's/# fr_FR.UTF-8 UTF-8/fr_FR.UTF-8 UTF-8/' /etc/locale.gen \
    && echo "LANG=${LANG}" > /etc/default/locale \
    && dpkg-reconfigure --frontend=noninteractive locales \
    && update-locale LANG=${LANG}

# Copie de la page web statique dans la racine du site Apache
COPY index.html /var/www/html/index.html

# Apache écoute sur le port 80 dans le conteneur
EXPOSE 80

# Lancer Apache2 au premier plan
CMD ["/usr/sbin/apache2ctl","-D","FOREGROUND"]
```

Construire l'image :

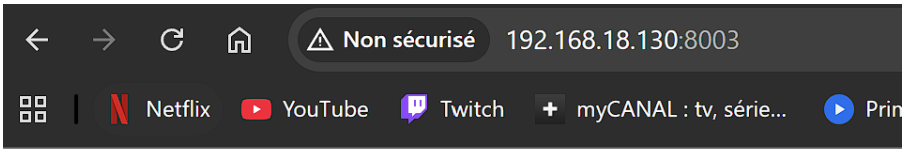
```
root@docker:~/contexte/debian12-apache2# docker build -t thomaska13/debian12-apache2:1.0 .
[+] Building 3.6s (10/10) FINISHED                                                                  docker:default
 => [internal] load build definition from dockerfile                                                          0.0s
 => => transferring dockerfile: 1.10kB                                                                        0.0s
 => [internal] load metadata for docker.io/library/debian:12                                                  2.0s
 => [auth] library/debian:pull token for registry-1.docker.io                                                 0.0s
 => [internal] load .dockerignore                                                                             0.0s
 => => transferring context: 2B                                                                               0.0s
 => [1/4] FROM docker.io/library/debian:12@sha256:7e5bc0e499a8d50cb1e32287944a90b9ec8fd7d500673e75daff3f52882f5  0.1s
 => => resolve docker.io/library/debian:12@sha256:7e5bc0e499a8d50cb1e32287944a90b9ec8fd7d500673e75daff3f52882f5  0.1s
 => [internal] load build context                                                                             0.0s
 => => transferring context: 252B                                                                             0.0s
 => CACHED [2/4] RUN apt update     && apt -y --no-install-recommends install         apache2         tzdata     0.0s
 => CACHED [3/4] RUN rm /etc/localtime     && ln -s /usr/share/zoneinfo/Europe/Paris /etc/localtime     && dpkg  0.0s
 => [4/4] COPY index.html /var/www/html/index.html                                                            0.2s
 => exporting to image                                                                                        0.8s
 => => exporting layers                                                                                       0.4s
 => => exporting manifest sha256:df67a5421e57d87c51083bd4042c3e3767f6fcc07614f064d6909cf2aad1aafe              0.1s
 => => exporting config sha256:18a603e04c117e34620b36ec41c39dedbea225ec12bd4e6617eb66d97b783ea4                0.0s
 => => exporting attestation manifest sha256:78f507662fd324f6889a901a8cbe9447dce0c9f54cc81b673f65ba68be328d43  0.1s
 => => exporting manifest list sha256:bc7055e68d77090110dfd04fcd9d18a021a673b7a7c898c8084d1bb352826ca7         0.0s
 => => naming to docker.io/thomaska13/debian12-apache2:1.0                                                    0.0s
 => => unpacking to docker.io/thomaska13/debian12-apache2:1.0                                                 0.1s
root@docker:~/contexte/debian12-apache2#
```

Lancer et tester le conteneur :

Lancement du conteneur :

```
root@docker:~/contexte/debian12-apache2# docker run -d --name debian12-web -p 8003:80 thomaska13/debian12-apache2:1.0
ef169fe62af2b9b83d283a7fbbd29e19317b5fff67ad6528c44753fadb7f466f
root@docker:~/contexte/debian12-apache2# docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS          PORTS
                                    NAMES
ef169fe62af2   thomaska13/debian12-apache2:1.0   "/usr/sbin/apache2ct…"   4 seconds ago    Up 3 seconds    0.0.0.0:800
3->80/tcp, [::]:8003->80/tcp   debian12-web
```

Teste dans un navigateur en tapant http://ip_de_ma_machine:8003 :



Test Apache2 sur Debian 12

Image Docker de thomaska13.

TP Docker : DÃ©ploiement d'un service

Se connecter à Docker Hub et publier :

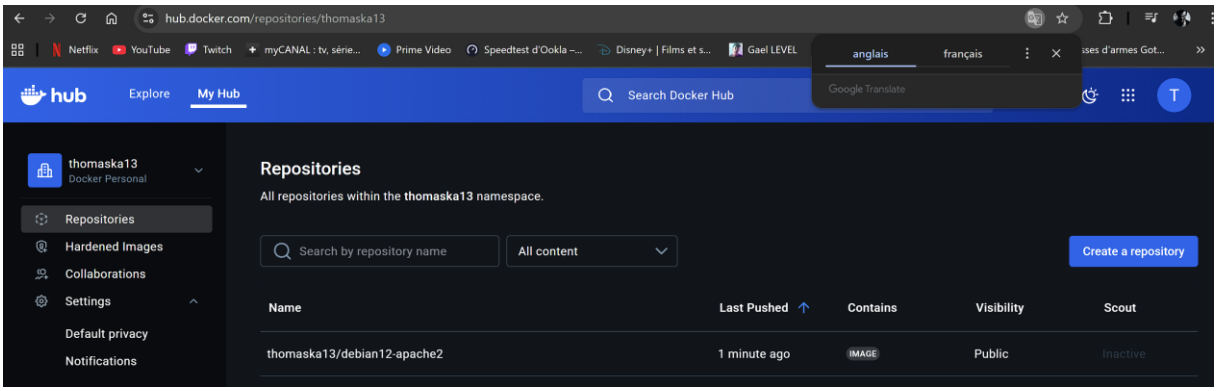
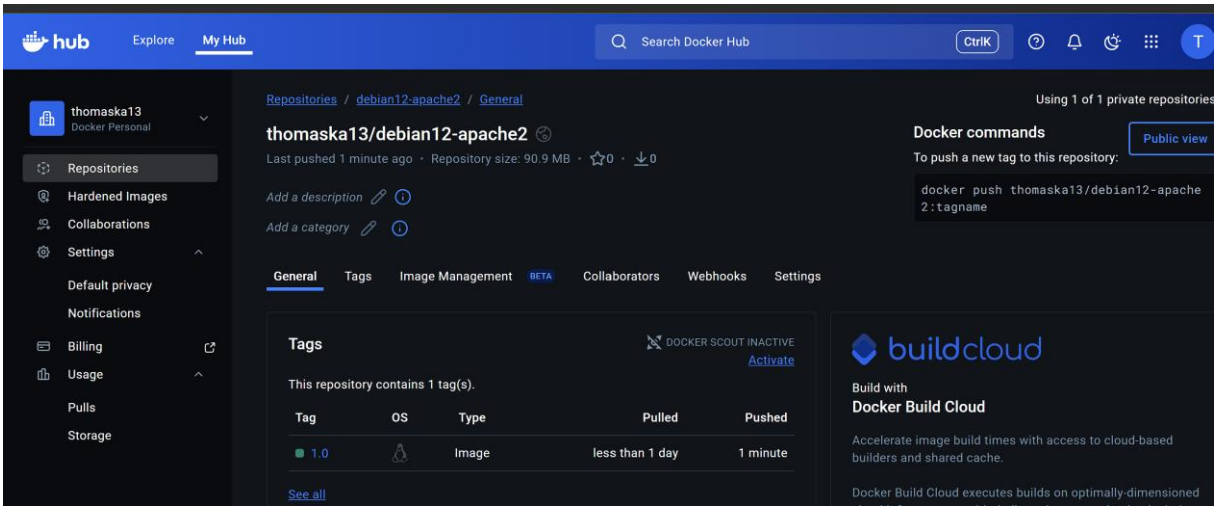
Ce connecter à docker hub en faisant la commande docker login -u nom_utilisateur :

```
root@docker:~/contexte/debian12-apache2# docker login -u thomaska13


i Info → A Personal Access Token (PAT) can be used instead.
         To create a PAT, visit https://app.docker.com/settings


Password:
Login Succeeded
```

Pousser l'image :

```
root@docker:~/contexte/debian12-apache2# docker push thomaska13/debian12-apache2:1.0
The push refers to repository [docker.io/thomaska13/debian12-apache2]
a3d3035cc3fc: Pushed
3b9c43831f34: Pushed
17dfab1b4f90: Pushed
4a8fa389d039: Pushed
c8443a297fa4: Pushed
1.0: digest: sha256:bc7055e68d77090110dfd04fcd9d18a021a673b7a7c898c8084d1bb352826ca7 size: 856
```

Vérifier depuis le site https://hub.docker.com :



En regardant dans le répertoire :